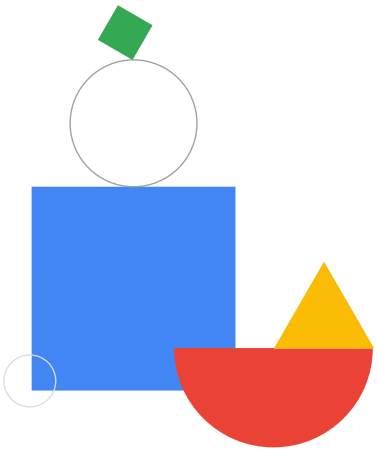
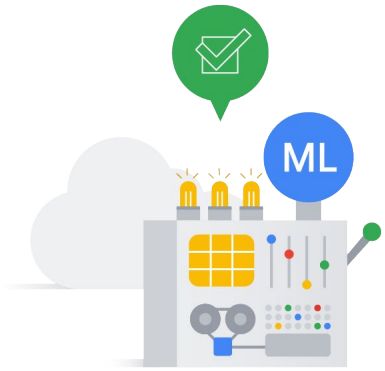
Machine Learning Operations (MLOps): Getting Started



Hi, my name is Esra Duygun and I am a Google Certified Professional Machine Learning Engineer and lead course developer.

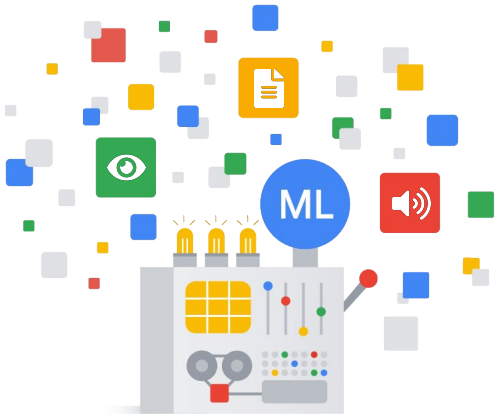
Welcome to Machine Learning Operations or MLOps Fundamentals, the first course of the new series of Machine Learning Operations topic.



How machine learning on Google Cloud can make tasks:

- Better
- Faster
- Easier

In the first series of courses, called Machine Learning on Google Cloud, you learn about how machine learning on Google Cloud can make tasks better, faster, and easier.

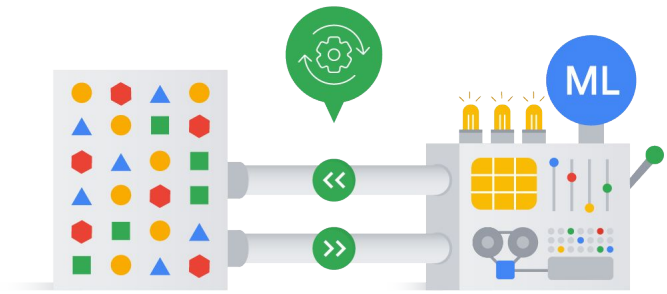


Focuses on more wide-ranging machine learning applications including:

- Computer vision
- Natural language processing
- Recommendation systems

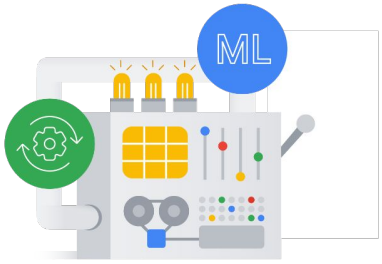
The second series of courses, **Advanced Machine Learning on Google Cloud**, focuses on more wide-ranging machine learning applications including:

- computer vision,
- natural language processing,
- and recommendation systems.



Machine learning models from an operational perspective

This series of courses is all about machine learning operations and focuses on machine learning models from an operational perspective.



Machine Learning
Operations (MLOps)

This particular course focuses on the MLOps concept and the considerations behind it.

So, what is MLOps?

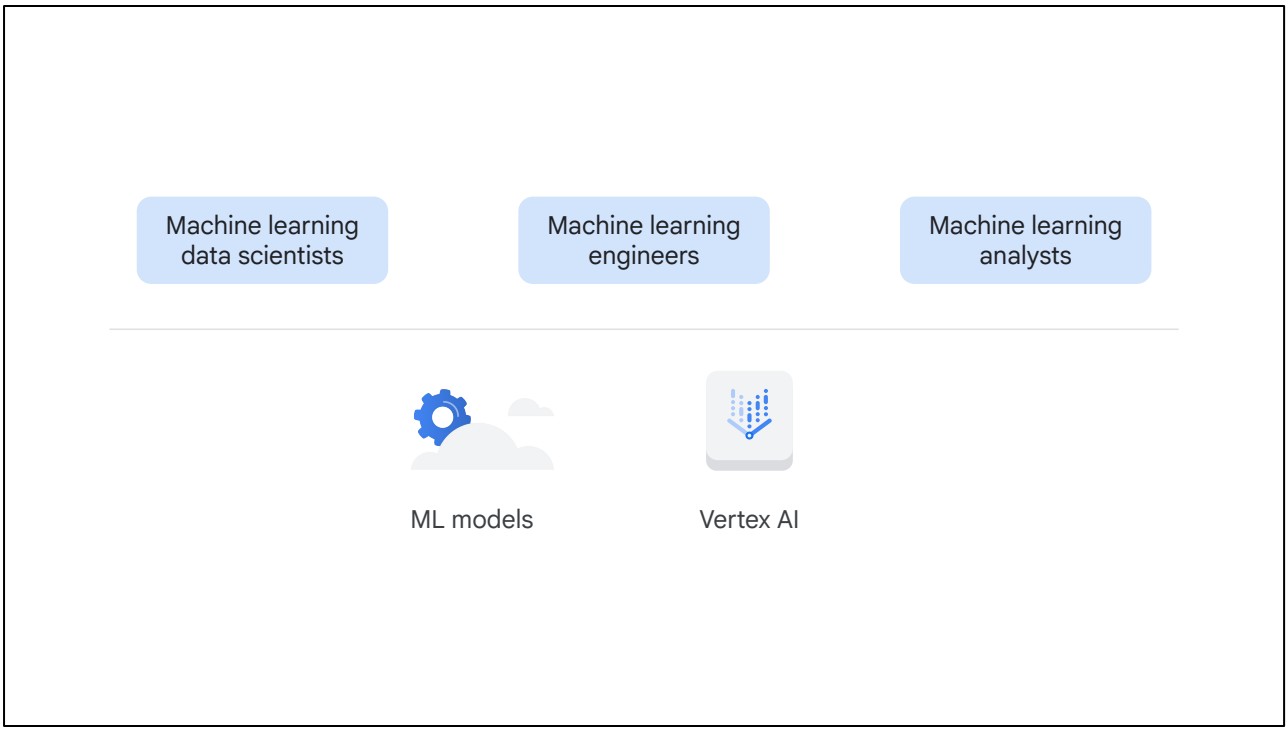
An ML engineering culture and practice that aims at:

- Unifying ML system development (Dev) and ML system operation (Ops).
- Guide teams through the challenges to the reproducibility of machine learning models.

MLOps is an ML engineering culture and practice that aims at unifying ML system development, or Dev, and ML system operations, or Ops, and guide teams through the challenges to the reproducibility of machine learning models.

MLOps takes both its name and some of the core principles and tools of DevOps, because the goals of MLOps and DevOps are the same: to shorten the development lifecycle of systems and ensure that high quality software is continuously developed, delivered, and maintained in production.

The unique challenges and needs that machine learning poses–managing the lifecycle of data, models, and code–have led MLOps to quickly evolve as a domain of its own.



This content is designed for dedicated or aspiring machine learning data scientists, engineers, and analysts who are interested in learning about machine learning in the cloud and using ML models and Vertex AI.

Prerequisites

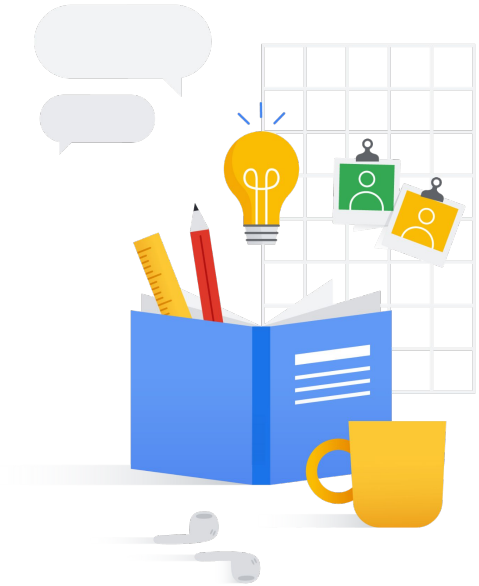
- Proficiency with Python on topics covered in the Crash Course on Python.
- Prior experience with foundational machine learning concepts and building machine learning solutions on Google Cloud as covered in the Machine Learning on Google Cloud course.

To get the most out of this specialization, it is recommended you have:

- Proficiency with Python on topics covered in the Crash Course on Python offered by Google, and
- Prior experience with foundational machine learning concepts and building machine learning solutions on Google Cloud as covered in the Machine Learning on Google Cloud courses.

Module 01

Employing Machine Learning Operations



In the first part of this course, Employing Machine Learning Operations, you explore machine learning models from an operational perspective.

- Examine the challenges that machine learning practitioners currently face.
- Get an introduction to the concept of DevOps in machine learning.
- Explore the machine learning lifecycle.

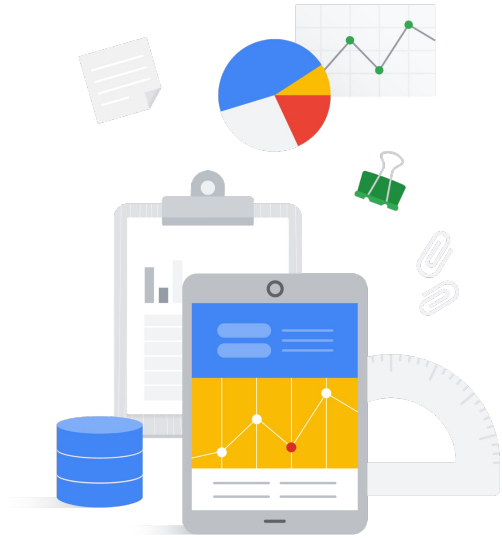
First, you examine the challenges that machine learning practitioners face when they operationalize ML models and make them available for production.

From there, you get an introduction to the concept of DevOps in machine learning.

And finally, you explore the machine learning lifecycle.

Module 02

What is Vertex AI and why does a unified platform matter?



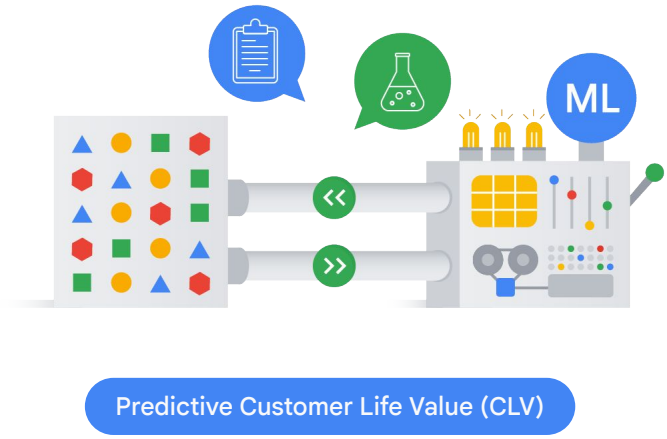
In the second part, What is Vertex AI and why does a unified platform matter?,

- Explore Vertex AI and learn why a unified platform matters.
- Discover MLOps capabilities of Vertex AI.
- Explore how Vertex AI helps with the MLOps workflow.

you learn about Vertex AI and its importance.

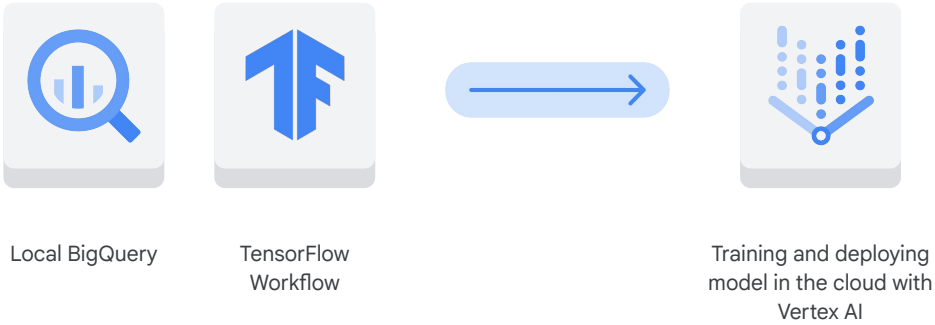
Then you discover the MLOps capabilities of Vertex AI.

Finally, you explore how Vertex AI helps with the MLOps workflow.

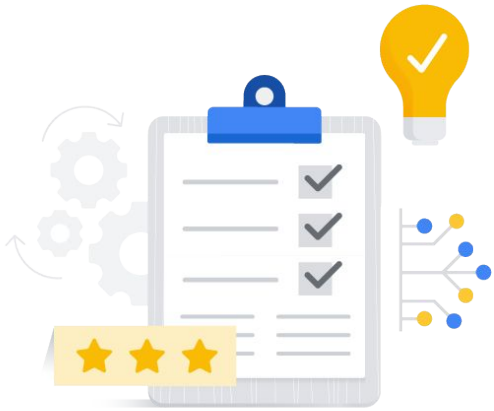


Through a hands-on lab at the end, you work on a high-value, real-world use case: predictive customer life value, or CLV in short.

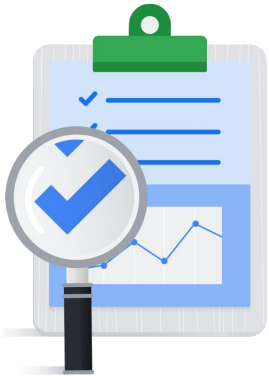
Predictive Customer Life Value (CLV)



You'll start with a local BigQuery and TensorFlow workflow that you might already be familiar with and progress toward training and deploying your model in the cloud with Vertex AI.



You also test your knowledge throughout the course with graded assessments.



ML projects through an operational lens

After you complete this course, you'll be able to see ML projects from an operational perspective.

You'll understand the concept of DevOps in ML and why you need to operationalize your ML models in a unified AI platform, like Vertex AI.

Enroll today at cloud.google.com/training

Enroll today to learn about machine learning operations.

END